
MATHEMATICAL BIOLOGY

How Does a Hydra Decide Where to Grow a Head?

Chemical and mechanical routes to pattern formation

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From regeneration to Turing instability, springs and energy



LECTURE MAP

The road from Hydra to pattern formation

From biological observations to mathematical mechanisms.

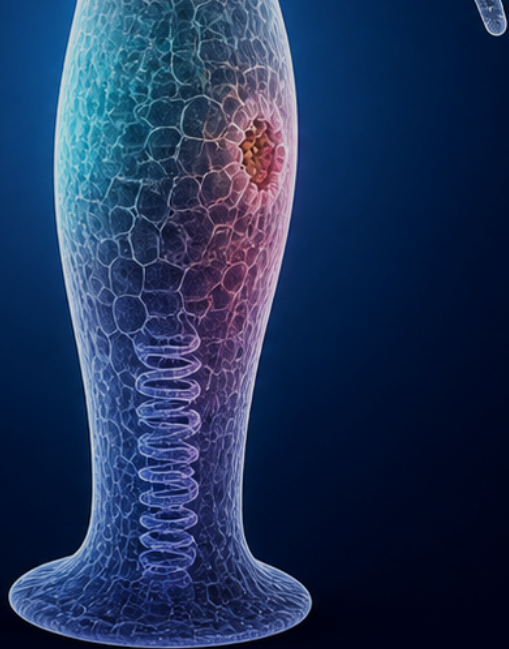
01 Mechanochemical pattern formation



01 MECHANOCHEMICAL PATTERN FORMATION

Why mechanics?

The classical inhibitor is not identified and mechanics offers an alternative long-range coupling



The activator has candidates but the inhibitor remains unclear

ACTIVATOR CANDIDATE

Wnt3

Stable localized Wnt3 expression marks the emerging head organizer

MISSING COMPONENT

?

No biochemical species has been established as the required fast long-range inhibitor

The mathematical inhibitor has a clear role but its biological identity is unresolved

Long-range inhibition need not be chemical

- **Chemical transport**

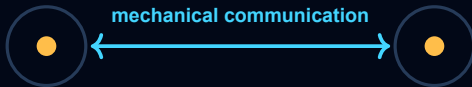
A molecule would have to transmit inhibition across the regenerating tissue

- **Mechanical coupling**

Stress and deformation can couple distant parts of a closed tissue without a second inhibitor

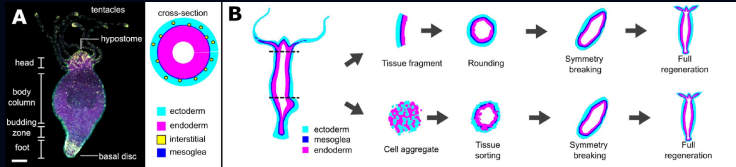
- **Modeling question**

Can one morphogen and tissue mechanics reproduce local activation and long-range inhibition?



a global constraint can transmit
local changes throughout the tissue

Regenerating spheroid



- **Initially symmetric**
A tissue fragment rounds into a hollow spheroid
- **Symmetry is broken**
One region becomes the future head organizer
- **A complete axis emerges**
The localized organizer directs whole-body regeneration

The central question

Can a local morphogen change the mechanics of the entire spheroid?

Can mechanics replace the diffusible inhibitor?

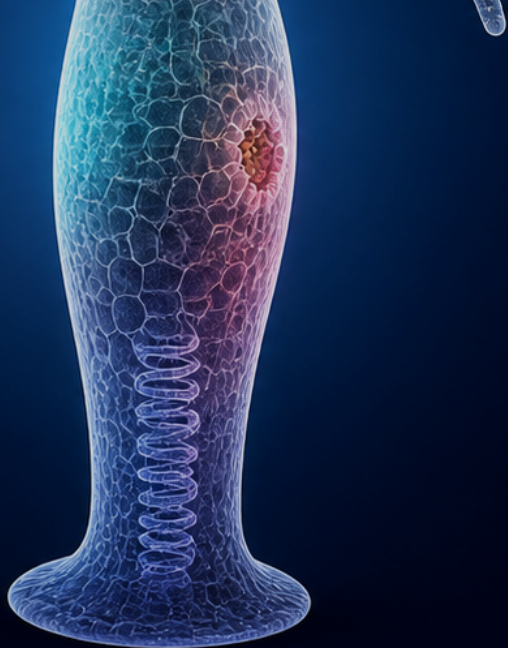


Build the simplest mechanochemical model that can answer this question

01 MECHANOCHEMICAL PATTERN FORMATION

Mechanics during regeneration

Swelling and rupture expose a two-way coupling between morphogen signaling and mechanics



One oscillation is a swelling and rupture cycle

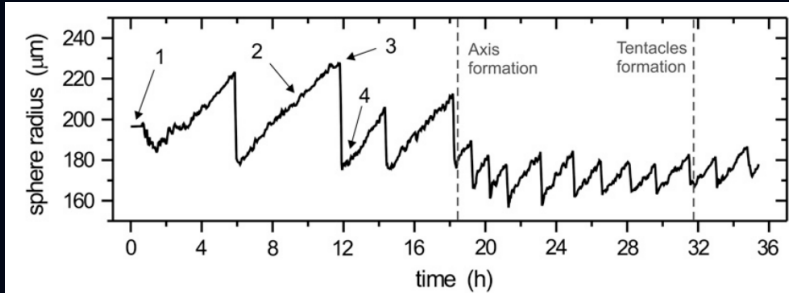
water enters → tissue stretches → tissue ruptures



the tissue deflates, heals, and the cycle begins again

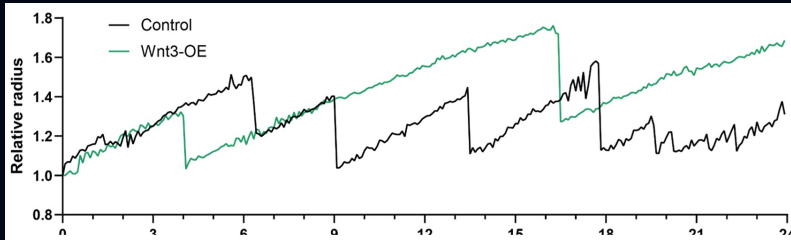
Osmotic pressure repeatedly converts fluid transport into tissue stretching

The spheroid radius records repeated rupture events



- **Slow increase**
Water influx gradually increases the spheroid radius
- **Rapid decrease**
Rupture releases fluid and abruptly reduces the radius
- **Changing dynamics**
The oscillation changes as the body axis and tentacles form

Wnt3 changes the oscillatory dynamics



- **Control**
Repeated rupture limits the gradual increase in radius
- **Wnt3 overexpression**
The timing and amplitude of rupture events are altered

Morphogen signaling and tissue-scale mechanics cannot be treated independently

Experiments suggest a mechanochemical coupling

TISSUE STRETCHING

⇓ Wnt3 production

Stretching correlates with organizer formation and Wnt3 expression

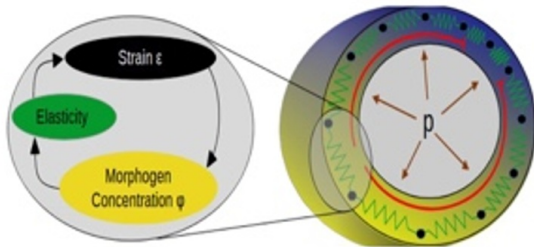
MORPHOGEN

⇓ Tissue elasticity

Morphogen signaling changes how strongly the tissue resists deformation

Each side of the feedback modifies the other

Mechanochemical feedback



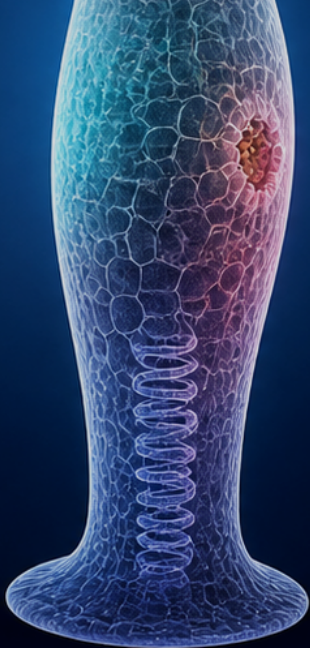
local activation long-range inhibition

- Local activation**
 Morphogen changes elasticity and concentrates strain locally
- Long-range inhibition**
 The closed pressurized tissue shares a finite total deformation
- Minimal ingredients**
 Morphogen dynamics, mechanics and one global constraint

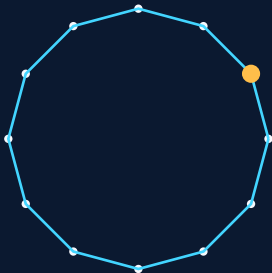
01 MECHANOCHEMICAL PATTERN FORMATION

From springs to a continuum

A closed ring of springs reveals how local elasticity produces a global redistribution of strain

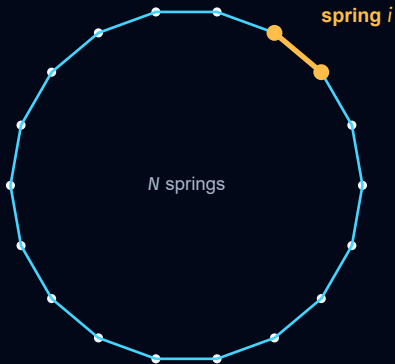


Local changes redistribute strain around the ring



- **Move one node**
A local displacement changes several neighboring springs
- **Change one stiffness**
A softer spring takes a larger share of the deformation
- **Close the ring**
The total extension cannot be assigned independently at every point

A ring contains many small springs



$$\Delta x = \frac{L}{N}$$

$$x_i = i\Delta x$$

$$i = 0, \dots, N - 1$$

Every spring represents a short material segment of reference length Δx

Compare the reference and deformed spring



extension of spring i : $\ell_i - \Delta x$

Strain measures relative extension

Absolute extension depends on the size of the spring

$$\varepsilon_i = \frac{\ell_i - \Delta x}{\Delta x}$$

- $\varepsilon_i = 0$
The spring keeps its reference length
- $\varepsilon_i > 0$
The spring is stretched
- $\varepsilon_i < 0$
The spring is compressed

The energy of one spring has the correct continuum scaling

Hooke's law assigns the spring energy

$$E_i = \frac{1}{2} k_i(u_i) (\ell_i - \Delta x)^2$$

$$k_i(u_i) = \frac{E(u_i)}{\Delta x}$$

$$E_i = \frac{1}{2} E(u_i) \varepsilon_i^2 \Delta x$$

- **Quadratic response**

Doubling the extension multiplies the elastic energy by four

- **Consistent subdivision**

Increasing the number of springs does not change the tissue stiffness

The factor Δx prepares the energy for a Riemann-sum limit

Morphogen concentration controls local stiffness

$$E'(u) < 0$$

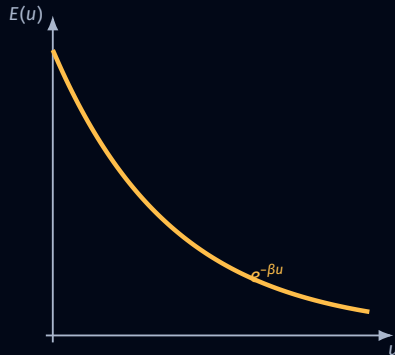
- **High morphogen**

The elastic modulus is smaller and the tissue is softer

- **Low morphogen**

The elastic modulus is larger and the tissue is stiffer

$$E(u) = e^{-\beta u} \quad \beta > 0$$



The local feedback is positive

higher u $\Rightarrow$ smaller $E(u)$ $\Rightarrow$ larger strain $\Rightarrow$ more u

- **Local activation**

A small local increase in morphogen makes the same region easier to stretch

- **Missing ingredient**

We still need to determine how the strain is distributed around the entire ring

The total discrete energy is a sum over all springs

Every spring contributes its local elastic energy

$$E_N = \sum_{i=0}^{N-1} E_i = \sum_{i=0}^{N-1} \frac{1}{2} E(u_i) \epsilon_i^2 \Delta x$$

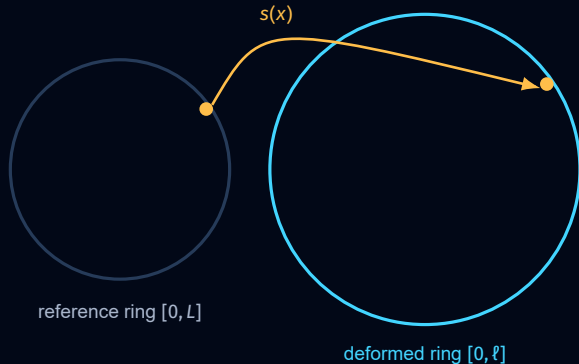
- **Local contribution**

Each term depends only on the strain and morphogen at spring i

- **Global configuration**

The total energy depends on how deformation is shared by all springs

A material coordinate follows the tissue



$$x \in [0, L]$$

$$s(x) \in [0, l]$$

- **Material label**
 x identifies a fixed piece of tissue
- **Deformed position**
 $s(x)$ gives its arc-length position after stretching

Discrete strain converges to a spatial derivative

$$\ell_i = s(x_{i+1}) - s(x_i)$$
$$\varepsilon_i = \frac{s(x_{i+1}) - s(x_i)}{\Delta x} - 1$$

$$\Delta x \longrightarrow 0 \quad \Rightarrow$$

$$\boxed{\varepsilon(x) = \partial_x s(x) - 1}$$

The strain compares the local deformed length with the reference length

The spring sum becomes a continuum energy

$$E_N = \sum_{i=0}^{N-1} \frac{1}{2} E(u_i) \varepsilon_i^2 \Delta x$$

$$N \longrightarrow \infty \quad \Downarrow$$

$$E_{\text{el}}[s; u] = \frac{1}{2} \int_0^L E(u(x)) \varepsilon(x)^2 dx$$

The discrete spring model has become a continuum elastic energy

The closed ring imposes a global length constraint

$$\begin{aligned}\ell - L &= \sum_{i=0}^{N-1} (\ell_i - \Delta x) \\ &= \sum_{i=0}^{N-1} \varepsilon_i \Delta x\end{aligned}$$



$$\ell - L = \int_0^L \varepsilon(x) dx$$

- **Total extension is prescribed**
Inflation determines how much longer the deformed ring is
- **Local strains are coupled**
More extension in one region leaves less extension for the rest of the ring

Mechanical equilibrium minimizes elastic energy

Mechanics relaxes rapidly compared with morphogen dynamics

$$s \in \arg \min \left\{ \frac{1}{2} \int_0^L E(u) \varepsilon^2 dx \right\}$$

$$\begin{aligned} \varepsilon &= \partial_x s - 1 \\ \int_0^L \varepsilon dx &= \ell - L \end{aligned}$$

- **Quasi-static mechanics**

For each current morphogen profile the tissue immediately reaches mechanical equilibrium

Perturb the deformation to compute the first variation

$$s_{\theta}(x) = s(x) + \theta h(x)$$

$$\begin{aligned}\varepsilon_{\theta} &= \partial_x s_{\theta} - 1 \\ &= \varepsilon + \theta \partial_x h\end{aligned}$$

The periodic perturbation h preserves the closed geometry

$$E_{\text{el}}[s_{\theta}; u] = \frac{1}{2} \int_0^L E(u) (\varepsilon + \theta \partial_x h)^2 dx$$

$$\begin{aligned}\delta E_{\text{el}} &= \left. \frac{d}{d\theta} E_{\text{el}}[s_{\theta}; u] \right|_{\theta=0} \\ &= \int_0^L E(u) \varepsilon \partial_x h dx\end{aligned}$$

Integration by parts gives mechanical equilibrium

$$\begin{aligned}\delta E_{\text{el}} &= \int_0^L E(u) \varepsilon \partial_x h \, dx \\ &= - \int_0^L \partial_x (E(u) \varepsilon) h \, dx\end{aligned}$$

The first variation vanishes for every admissible perturbation h

$$\partial_x (E(u) \varepsilon) = 0$$

Mechanical equilibrium determines the strain explicitly

$$E(u(x, t))\varepsilon(x, t) = \sigma(t)$$

$$\varepsilon(x, t) = \frac{\sigma(t)}{E(u(x, t))}$$

$$\ell - L = \sigma(t) \int_0^L \frac{1}{E(u(y, t))} dy$$

$$\varepsilon(x, t) = \frac{\ell - L}{E(u(x, t)) \int_0^L \frac{1}{E(u(y, t))} dy}$$

The full model couples dynamics and mechanical equilibrium

$$\partial_t u = D \partial_{xx} u - \alpha u + \kappa \varepsilon$$

$$0 = \partial_x (E(u) \varepsilon)$$

$$\ell - L = \int_0^L \varepsilon(x, t) dx$$

supplemented with periodic boundary conditions and an initial morphogen profile

- **One evolution equation**
Only the morphogen has its own time dynamics
- **Instantaneous mechanics**
Strain is determined by equilibrium at every time
- **One global constraint**
The prescribed total extension couples the entire ring

Mechanical variables can be eliminated exactly

$$\partial_t u = D \partial_{xx} u - \alpha u + \kappa \varepsilon$$

$$\varepsilon(x, t) = \frac{\ell - L}{E(u(x, t)) \int_0^L \frac{1}{E(u(y, t))} dy}$$



$$\partial_t u = D \partial_{xx} u - \alpha u + \kappa \frac{\ell - L}{E(u(x, t)) \int_0^L \frac{1}{E(u(y, t))} dy}$$

Exponential softening produces the nonlocal feedback

$$E(u) = e^{-\beta u}$$

$$\frac{1}{E(u)} = e^{\beta u}$$

$$\varepsilon(x, t) = (\ell - L) \frac{e^{\beta u(x, t)}}{\int_0^L e^{\beta u(y, t)} dy}$$

- Local factor**

$e^{\beta u(x, t)}$ amplifies regions with larger morphogen

- Global factor**

The denominator preserves the prescribed total extension

The mechanics generates a local exponential activation divided by a global normalization

Nondimensionalization gives one nonlocal equation

Natural variables

$$\tilde{x} = \frac{x}{L}$$

$$\tilde{t} = \alpha t$$

$$\tilde{u} = \beta u$$

$$\partial_t u = D \partial_{xx} u - u + \kappa \frac{e^{u(x,t)}}{\int_0^1 e^{u(y,t)} dy}$$

The tildes are dropped and the domain is the unit circle $\mathbb{T} = [0, 1]$

$$u(0, t) = u(1, t)$$

$$\partial_x u(0, t) = \partial_x u(1, t)$$

$$u(x, 0) = u_0(x)$$

One equation realizes the activator–inhibitor paradigm

LOCAL ACTIVATION

$$e^{u(x,t)}$$

A region with more morphogen becomes softer and receives more strain-driven production

LONG-RANGE INHIBITION

$$\int_0^1 e^{u(y,t)} dy$$

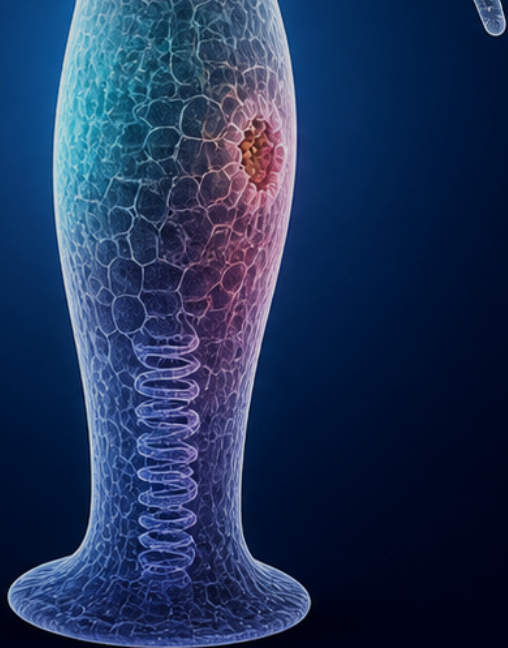
Increased production in one region reduces the share of extension available elsewhere

Mechanical equilibrium converts a local positive feedback into global competition

01 MECHANOCHEMICAL PATTERN FORMATION

Steady states and stability

Variational structure determines when patterns exist and which stationary profiles can be stable



Stationary patterns solve a nonlocal boundary-value problem

$$0 = D\partial_{xx}U - U + \kappa \frac{e^{U(x)}}{\int_0^1 e^{U(y)} dy}$$

$$U(0) = U(1)$$

$$\partial_x U(0) = \partial_x U(1)$$

- **Homogeneous solution**
 $\bar{U} \equiv \kappa$ is always a stationary solution

- **Patterned solution**
 A nonconstant periodic function $U(x)$ represents a spatial pattern

Every stationary solution has mean value $\int_0^1 U(x) dx = \kappa$

Small diffusion permits patterns while large diffusion suppresses them

THEOREM

For every $\kappa > 0$ there exist constants

$$0 < D_{\min}(\kappa) < D_{\max}(\kappa)$$

- $0 < D < D_{\min}(\kappa)$

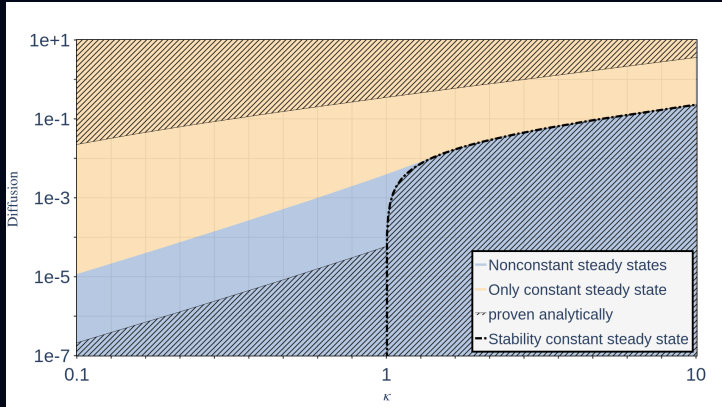
A nonconstant stationary solution exists and is Lyapunov stable in L^2

- $D > D_{\max}(\kappa)$

$\bar{U} \equiv \kappa$ is the unique stationary solution and is globally asymptotically stable

The analytic thresholds are sufficient bounds and need not be optimal

The parameter plane separates homogeneous and patterned regimes



- **Patterns**
Nonconstant steady states are found for sufficiently small diffusion
- **Homogeneous state**
Large diffusion removes spatial variation

Hatched regions are established analytically

Linearization asks whether a stationary profile amplifies perturbations

$$u(x, t) = U(x) + e^{\nu t} \varphi(x)$$

Keeping only first-order terms gives the nonlocal eigenvalue problem

$$D \partial_{xx} \varphi + \left(\kappa \frac{e^U}{\int_0^1 e^U} - 1 - \nu \right) \varphi = \kappa \frac{e^U}{\left(\int_0^1 e^U \right)^2} \int_0^1 e^U \varphi$$

- $\nu < 0$

The corresponding perturbation decays

- $\nu > 0$

The corresponding perturbation grows

A local problem helps locate the nonlocal eigenvalues

NONLOCAL PROBLEM

$$D\partial_{xx}\varphi + (A - \nu)\varphi = MC(x) \int_0^1 C(y)\varphi(y) dy$$

Every value of φ is coupled through the integral

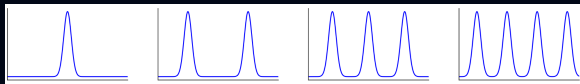
LOCAL PROBLEM

$$D\partial_{xx}\psi + (A - \lambda)\psi = 0$$

Classical oscillation theory orders the eigenfunctions by their zeros

Relevant nonlocal eigenvalues lie between eigenvalues of the associated local problem

Stable mechanochemical patterns are necessarily unimodal



THEOREM

- **Stable nonconstant solution**

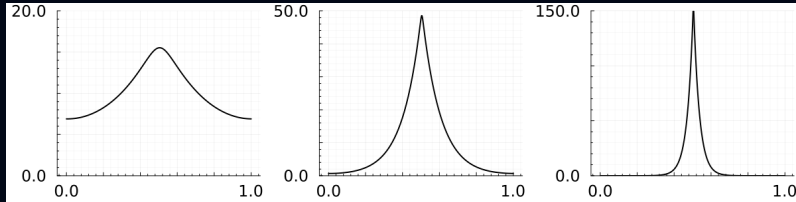
The stable solution obtained by energy minimization has one peak

- **Every multimodal solution**

All solutions with two or more peaks are linearly and nonlinearly unstable

The global mechanical constraint strongly selects a single organizer

Diffusion changes the width but not the number of stable peaks

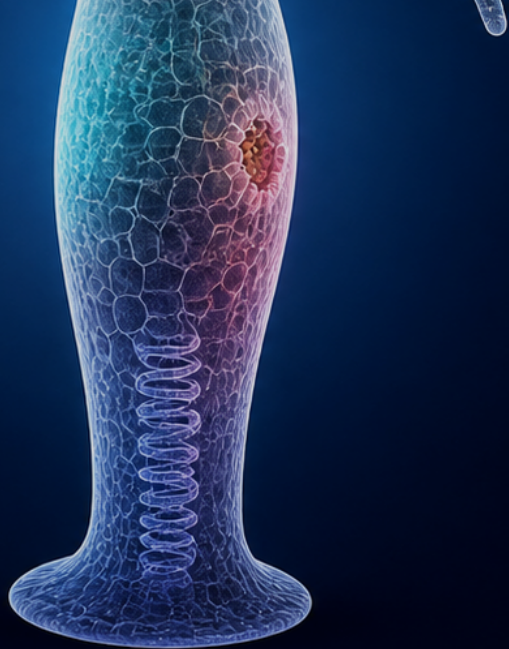


- **Smaller diffusion**
The stable peak becomes sharper and more localized
- **Larger diffusion**
The stable profile becomes broader
- **Robust selection**
The stable pattern retains a single dominant peak

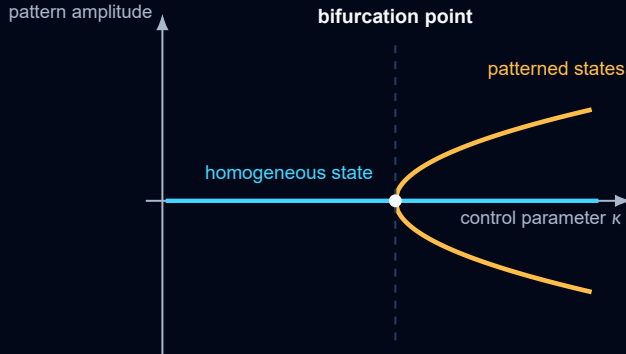
01 MECHANOCHEMICAL PATTERN FORMATION

Bifurcations and oscillations

Stationary branches emerge when a homogeneous state changes stability and can generate tissue oscillations



A bifurcation marks a qualitative change in stationary states



- **Before the threshold**
Only the homogeneous stationary state is visible nearby
- **At the threshold**
A spatial mode becomes neutral and a patterned branch can emerge
- **After the threshold**
The number or stability of stationary states has changed

The homogeneous state loses stability at discrete thresholds

$$\bar{U} \equiv \kappa$$

$$\varphi_n(x) = \cos(2\pi nx)$$

Linearization assigns one growth rate to every spatial mode

$$\nu_n = \kappa - 1 - 4\pi^2 n^2 D$$

$$\kappa_n = 1 + 4\pi^2 n^2 D$$

- $n = 1$

The first nonconstant mode has one maximum and one minimum

- $n = 2$

The next mode produces a two-peak spatial structure

- **Discrete thresholds**

Each mode changes stability at its own value of κ

The first patterned branch can be subcritical or supercritical

THEOREM

At $\kappa = \kappa_n$ a branch of nonconstant stationary solutions bifurcates from the homogeneous state $\bar{U} \equiv \kappa$

- **Subcritical branch**

$\kappa_n < 3/2$ equivalently $D < (8\pi^2 n^2)^{-1}$

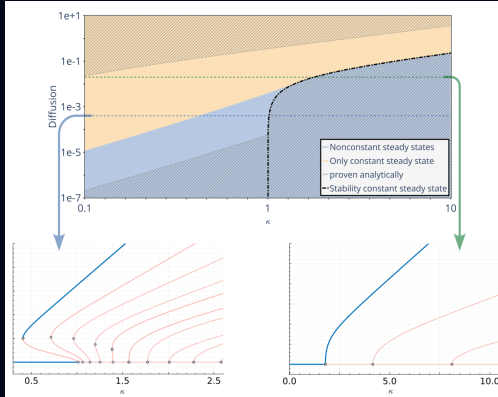
- **Supercritical branch**

$\kappa_n > 3/2$ equivalently $D > (8\pi^2 n^2)^{-1}$

Near the bifurcation only the supercritical branch generated by the first mode can be stable

The direction of the branch determines whether small patterned states appear before or after the linear threshold

Diffusion determines the type of each bifurcation



- **Small diffusion**
The branch can bend toward smaller values of κ
- **Larger diffusion**
The branch can emerge toward larger values of κ

Each spatial mode has its own transition curve

Tissue length turns the stationary picture into dynamics

Let $\ell(t)$ denote the current circumference and L its reference value

$$\partial_t u = D \partial_{xx} u - u + \kappa (\ell(t) - L) \frac{e^{u(x,t)}}{\int_0^1 e^{u(y,t)} dy}$$

- **Morphogen rises**

The tissue softens and the mechanical equilibrium shifts

- **The spheroid expands**

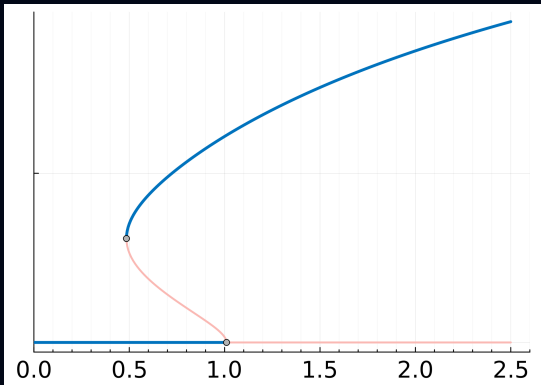
Increased strain strengthens morphogen production

- **Feedback relaxes**

Degradation and mechanical restoration can reverse the expansion

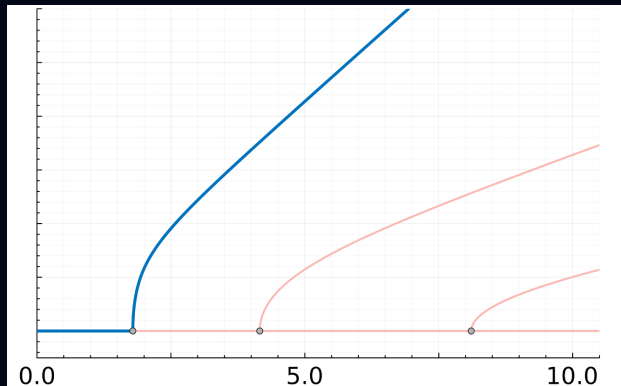
Coupling pattern formation to tissue size creates a route from static symmetry breaking to oscillations

Subcritical feedback permits abrupt switching



- **Coexisting states**
A homogeneous and a patterned state may be available for the same parameter value
- **Sudden transition**
Crossing a fold can force the system to jump to a distant branch

Supercritical feedback produces a gradual patterned response



- **Continuous onset**
The patterned amplitude grows continuously from the homogeneous state
- **Smooth mechanical response**
Tissue deformation can track the changing stationary branch

Mechanics can realize an activator–inhibitor mechanism without a chemical inhibitor

- **Local positive feedback**

Stretch increases morphogen production and morphogen softens the tissue

- **Global mechanical competition**

The fixed total length couples every point through one nonlocal constraint

- **Single organizer**

Stable nonconstant stationary states contain one dominant peak

morphogen $\Rightarrow$ softening $\Rightarrow$ strain $\Rightarrow$ morphogen

A single morphogen equation coupled to mechanics can implement local activation and long-range inhibition

Thank you

for your attention

